Po-Wei (Brian) Chen

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RESEARCH INTERESTS

Statistical Machine Learning, Probabilistic Generative Models, Bayesian Statistics.

EDUCATION

Soochow University, Taipei, Taiwan
B.B.A. in Financial Engineering and Actuarial Mathematics — Sept. 2022 – June 2025

- Graduated one year early (met criteria of full credit completion and top 10% class rank each semester)
- GPA 4.0/4.0; Average 95.2/100; Class Rank 1^{st}/53
- Relevant Coursework: *Advanced Calculus*, *Calculus(I)(II)*, *Linear Algebra(I)(II)*, *Probability and Statistics(I)(II)*, *Advanced Statistics*, *Regression Analysis*, *Statistical Inference*, *Applied Statistics*, *Numerical Analysis*

WORKING EXPERIENCE

Research Assistant (Full-time)
Institute of Statistical Science, Academia Sinica, the leading research institute in Taiwan — Aug. 2025 – present
PI: Prof. Hsin-Cheng Huang, Distinguished Research Fellow

- Augmented diffusion-based generative models with catalytic priors that incorporate information from simpler statistical structures to guide the training process.
- Assisted in conference operations for the Joint Meetings of the 2025 Taipei International Statistical Symposium and the 13th ICSA International Conference.

High School Mathematics Tutor — Aug. 2024 – June 2025
Taiwan TACT Education (Private Tutoring Center), Minsheng Branch, Taipei, Taiwan

- Taught high school mathematics and prepared students for university entrance examinations.

CONFERENCE PRESENTATIONS

- 2026, "Bayesian Inference for Diffusion Models: Integrating Synthetic and Observed Data with Catalytic Priors" *ISI-ISM-ISSAS Joint Conference*, Academia Sinica, Taipei, Taiwan. [Poster Presentation]
  - This work was later retitled "Bridging Synthetic and Real Data with Catalytic Priors for Diffusion Models."
- 2025, "Empirical Asset Pricing via Machine Learning for Cryptocurrency" *Annual Conference and International Symposium of the Taiwan Risk and Insurance Association*, Taipei, Taiwan. [20-Minute Oral Presentation]

HONORS & AWARDS

2026 ISI-ISM-ISSAS Joint Conference Poster Competition Award
Group B: Computational Statistics, Machine Learning & Optimization

- Title: Bayesian Inference for Diffusion Models: Integrating Synthetic and Observed Data with Catalytic Priors.

Outstanding Graduating Student
Soochow University, Taipei, Taiwan

- Recognized for graduating one year early and achieving the highest cumulative academic ranking in the class.

Academic Excellence Award (Fall 2022–Spring 2025; awarded every semester)
Soochow University, Taipei, Taiwan

- Awarded to the top three students in the class each semester.

RESEARCH PROJECTS

Bridging Synthetic and Real Data with Catalytic Priors for Diffusion Models
Supervisor: Prof. Hsin-Cheng Huang

- Developed a novel diffusion-based framework augmented with catalytic priors that leverage simpler statistical structures to guide learning in data-scarce settings. By adding a prior-based regularization term to enhances the model's generalization ability.

Empirical Asset Pricing via Machine Learning for Cryptocurrency
Supervisor: Prof. Hung-Wen Cheng

- Compared various machine learning models for factor-based quantile investment strategies and demonstrated the effectiveness of machine learning in cryptocurrency factor investing, where tree-based methods and Support Vector Regression achieved strong and stable returns in finer quantile portfolios, yielding approximately a 60% win rate over the market benchmark.

Regression Analysis for Crime Rate in New York State (course project) [GitHub page]
Course: Regression Analysis — Feb. 2025 – June 2025

- Applied multiple linear regression to analyze crime rates across 62 New York State counties; model achieved strong overall significance ($F\ statistic: 33.47$, $p < 0.001$) and Adjusted $R^2 = 0.6149$, identifying median income, education level, and population size as highly significant determinants (all $p < 0.001$).

Factor Investing with Machine Learning (course project) [GitHub page]
Course: Big Data and Fintech Practice — Feb. 2024 – June 2024

- Implemented OLS, Random Forest, and Neural Network models to predict future information coefficients between factor values and subsequent stock returns, and constructed investment strategies based on the predictions. Random Forest substantially outperformed the market benchmark, achieving approximately a 60% win rate across all backtested portfolios.

CERTIFICATIONS

Open Online Courses

- Python for Everybody Specialization, University of Michigan (via Coursera)
- Programming for Business Computing in Python (1), National Taiwan University (via Coursera)

Exam P (Probability), Society of Actuaries (SOA)

EXTRACURRICULAR ACTIVITIES

2025 Statistics Cup Basketball Tournament Champions — Nov. 2025
Institute of Statistical Science, Academia Sinica, Taipei, Taiwan

Vice Captain of Basketball Team — Sept. 2024 – June 2025
Department of Financial Engineering and Actuarial Mathematics, Soochow University, Taipei, Taiwan

- Led the team to six championships in inter-departmental competitions.

SKILLS & LANGUAGES

Programming: Python (Numpy, Pandas, Matplotlib), R (Markdown, Shiny), C++, SQL
Machine Learning Packages: PyTorch, TensorFlow, scikit-learn
Languages: Mandarin (native), English (TOEFL: 97/120, R: 27, L: 24, S: 24, W: 22)
Hobbies: Basketball, Photograph